ST09

Computers, IT and ICT

Roughly 3-5 questions a paper, and the Rajasthan e-governance part alone has supplied a question in 2016, 2018, 2021 and 2024. Networks, data units and the new technology missions are the other reliable suppliers. Nothing here needs technical depth - it needs exact names, years and ministries.

This is the largest chapter in the Science and Technology block and it has changed character. Ten years ago RPSC asked what a byte is and which key saves a file. It still asks that - but it now also asks about the National Quantum Mission, the IndiaAI Mission, the DPDP Act and Rajasthan's own portals. So you have two jobs: keep the old basics ready for the easy marks, and learn the new missions with their years, outlays and parent ministries.

Computers, IT and ICT

Generations

- I vacuum tube (1946-59)
- II transistor (1959-65)
- III integrated circuit (1965-71)
- IV microprocessor, VLSI (1971-80)
- V AI, ULSI (1980 onwards)

Hardware

- CPU = ALU + Control Unit + registers
- Registers > cache > RAM > secondary
- RAM volatile, ROM not
- EPROM - erased by UV light
- MICR on cheques; plotter is OUTPUT

Data and number systems

- nibble 4 bits, byte 8 bits
- KB 1024 bytes, then MB GB TB PB
- Binary 2, octal 8, decimal 10, hex 16
- ASCII 7-bit, 128 characters
- Unicode is the universal standard

Software

- System vs application software
- Compiler all at once, interpreter line by line
- 1GL machine to 5GL LISP/Prolog
- FORTRAN 1957 - first high-level
- Codd - relational model, 1970

Networks

- PAN < LAN < MAN < WAN
- Bus, star, ring, mesh, tree, hybrid
- OSI 7 layers, TCP/IP 4
- IPv4 32-bit, IPv6 128-bit
- HTTP 80, HTTPS 443, SMTP 25, DNS 53

Internet and cloud

- ARPANET 1969
- WWW - Berners-Lee, CERN, 1989
- India commercial internet 15 Aug 1995, VSNL
- IaaS, PaaS, SaaS
- MeghRaj - GoI cloud

New technologies

- IoT, big data - five Vs
- Blockchain - distributed, immutable
- AI coined 1956, McCarthy, Dartmouth
- Transformer 2017 - basis of generative AI
- Qubit, superposition, entanglement

India's missions

- IndiaAI Mission 2024, Rs 10,372 cr, MeitY
- National Quantum Mission 2023, Rs 6,003.65 cr, DST
- 5G launched 1 October 2022
- NSM 2015 - PARAM Shivay, PARAM Rudra, AIRAWAT

Cyber security and law

- Virus needs a host, worm spreads itself
- IT Act 2000 - UNCITRAL model law
- Sec 66A struck down - Shreya Singhal 2015
- CERT-In under MeitY; I4C under MHA, 1930
- DPDP Act 2023 - max penalty Rs 250 crore

E-governance

- Digital India - 1 July 2015
- DigiLocker, UMANG, Aadhaar, UPI, ONDC

–Kerala - first digital state, 2016
–e-Mitra, Jan Aadhaar, Raj Kaj, iStart
–DoIT&C is Rajasthan's nodal department

Generations of computers - the matching question you can bank

Every generation is named after the electronic component it was built from. That is the whole logic, and that is exactly how RPSC asks it - generation on the left, component on the right. Learn the five components in order and you have the answer for any arrangement of the pairs.

Five generations by principal component

I (1946-59) - Vacuum tube → II (1959-65) - Transistor → III (1965-71) - Integrated circuit → IV (1971-80) - Microprocessor and VLSI → V (1980 onwards) - Artificial intelligence, ULSI, parallel processing

- Charles Babbage is the 'Father of the Computer' - Difference Engine 1822, Analytical Engine 1837.
- Ada Lovelace is regarded as the world's first programmer.
- ENIAC (1946) was the first general-purpose electronic digital computer.
- UNIVAC-I was the first computer sold commercially.
- EDVAC embodied the stored-programme concept, associated with John von Neumann.

YAAD RAKHO

Say the components as one string: Vacuum - Transistor - IC - Microprocessor - AI. Five words, five generations, in order. The years are secondary.

Hardware – the CPU and the memory hierarchy

The CPU is made of three things: the Arithmetic Logic Unit, which does the arithmetic and the logic; the Control Unit, which fetches and decodes instructions and directs everybody else; and registers, which are tiny holding places inside the processor. Registers are the fastest memory in the machine, not the slowest - RPSC has built a statement question on exactly that reversal.

Memory hierarchy – fastest and smallest at the top

Registers - inside the CPU, fastest, smallest → Cache - faster than RAM, small capacity → Main memory (RAM) - the working memory → Secondary storage - hard disk, SSD, optical, tape

Types of memory – the volatile test

Memory	Volatile?	Key point
RAM	Yes	Working memory; contents lost when power goes off
ROM	No	Retains contents without power; holds firmware
PROM	No	Programmable once, then permanent
EPROM	No	Contents erased by exposure to ultraviolet light
EEPROM	No	Erased electrically
Flash memory	No	A form of EEPROM - pen drives, memory cards, SSDs
Cache	Yes	Faster than main memory but much smaller

- Input devices: keyboard, mouse, scanner, joystick, light pen, barcode reader, OMR, OCR and MICR.
- MICR - Magnetic Ink Character Recognition - is the one used on bank cheques.
- Output devices: monitor, printer, plotter, speaker.
- Printers: dot-matrix is an impact printer; laser and inkjet are non-impact.
- As you go down the hierarchy speed falls and capacity rises - and cost per bit falls.

TRAP

Two reversals RPSC loves. First: a plotter is an OUTPUT device, even though it appears in lists of input devices as the odd one out. Second: the CPU cannot execute instructions lying on a hard disk - they must first be loaded into main memory. A statement saying otherwise is false.

Number systems and data units

Computers store everything as 0 and 1. Number systems are simply different bases for writing the same value, and RPSC asks two things only - the base of a system, and the order of the data units. Both are memory work. Nothing here requires calculation beyond a small binary conversion.

The four number systems

System	Base	Digits used
Binary	2	0, 1
Octal	8	0 to 7
Decimal	10	0 to 9
Hexadecimal	16	0 to 9 and A to F

Data units in ascending order

Unit	Equals
Bit	A single 0 or 1
Nibble	4 bits
Byte	8 bits
Kilobyte (KB)	1024 bytes
Megabyte (MB)	1024 KB
Gigabyte (GB)	1024 MB
Terabyte (TB)	1024 GB
Then PB, EB, ZB, YB	each 1024 times the one before

- A nibble is half a byte - four bits. That single fact is a whole question.
- In hexadecimal, A stands for 10 and F for 15.
- Decimal 25 in binary is 11001. Convert by repeated division by 2 and reading the remainders upwards.
- ASCII is a 7-bit code covering 128 characters; Unicode is the universal standard that covers the world's scripts, including Devanagari.
- Ascending order to memorise: KB, MB, GB, TB, PB, EB, ZB, YB.

Software, operating systems and programming languages

Software splits in two. System software runs the machine - the operating system, compilers, assemblers, interpreters, device drivers. Application software does your work - a word processor, a browser, a payroll package. The operating system is the manager sitting between hardware and everything else: it handles memory, files, processes and devices.

- Operating systems: Windows, Linux, Unix, macOS, Android, iOS.
- Android is built on the Linux kernel.

- Oracle is a database management system, NOT an operating system - the classic odd-one-out option.
- Open source means the source code is freely available to read, modify and redistribute; Linux is the standard example.
- Firmware sits in ROM and starts the machine; a device driver lets the OS talk to a particular piece of hardware.
- MS Office is application software. Shortcuts asked as odd-one-out: Ctrl+C copy, Ctrl+X cut, Ctrl+V paste, Ctrl+Z undo, Ctrl+Y redo, Ctrl+S save, Ctrl+P print, Ctrl+A select all, Ctrl+F find.
- Function keys: F1 help, F2 rename, F5 refresh, Alt+F4 close the window.
- File extensions get asked the same way: .docx word processing, .xlsx spreadsheet, .pptx presentation, .mp3 audio, .jpg image, .pdf portable document.

- A compiler translates the whole programme at one go and reports all the errors together.
- An interpreter translates and executes line by line, stopping at the first error.
- An assembler converts assembly language into machine code.
- DBMS: E.F. Codd proposed the relational model in 1970.
- In a relation, a row is a tuple or record and a column is an attribute or field; the primary key uniquely identifies a row.

Generations of programming languages

Generation	Type	Examples
1GL	Machine language	Binary instructions the CPU runs directly
2GL	Assembly language	Mnemonics such as ADD, MOV
3GL	High-level languages	FORTRAN (1957, the first), COBOL, C, Java, Python
4GL	Query and report languages	SQL
5GL	Artificial intelligence languages	LISP, Prolog

Computer networks – the part RPSC has started leaning on

A network is just computers joined so they can share data and resources. Three things get asked: the classification by geographical span, the topology (the shape of the wiring), and the protocol-to-function matching. Take them in that order and the section is short.

Networks by geographical span – smallest to largest

Network	Span	Typical example
PAN – Personal Area Network	A few metres around one person	A Bluetooth link between phone and earphones
LAN – Local Area Network	One building, office or campus	An office or college network
MAN – Metropolitan Area Network	A city	A city-wide cable or broadband network
WAN – Wide Area Network	A country or the whole world	The internet itself

Network topologies

Topology	Layout	Note
Bus	All nodes share a single backbone cable	Cheapest; failure of the backbone brings the whole network down
Star	Every node connects to a central hub or switch	Easy to manage; failure of the hub kills the network
Ring	Nodes form a closed loop, data moves node to node	A break in the loop disrupts the network
Mesh	Every node connected to every other node	Most fault-tolerant, but needs the most cabling
Tree	Hierarchical, star networks joined to a backbone	Used in large structured setups
Hybrid	Any combination of the above	Most real networks

OSI reference model versus TCP/IP model

OSI reference model	TCP/IP model
7 layers	4 layers
Physical, Data Link, Network, Transport, Session, Presentation, Application	Network Access, Internet, Transport, Application
A theoretical reference framework	The model the internet actually runs on
Layers are strictly separated	Layers are merged and practical

- Devices by layer: repeater and hub work at the physical layer; switch and bridge at the data link layer; a router at the network layer.
- A gateway joins two dissimilar networks. A modem modulates and demodulates – digital to analogue and back.
- An IPv4 address is 32 bits, written as four octets in dotted decimal, for example 192.168.1.1.
- An IPv6 address is 128 bits – created because IPv4 addresses ran out.
- Bandwidth is the data-carrying capacity of a link, measured in bits per second.

Common protocols and their default ports

Protocol	Function	Default port
HTTP	Transfer of web pages	80
HTTPS	Secure (encrypted) web transfer	443
FTP	File transfer between hosts	20 and 21
SMTP	Sending electronic mail	25
POP3	Retrieving mail from a server	110
DNS	Translating domain names into IP addresses	53
Telnet	Remote login	23
DHCP	Automatic assignment of IP addresses to hosts	–

The internet, the web and cloud computing

Keep the two apart. The internet is the global network of networks - the wires, the routers, the addressing. The World Wide Web is one service running on it, a system of linked pages accessed through a browser. Email, file transfer and video calls also run on the internet without being 'the web'. RPSC has asked the invention of the web separately from the origin of the internet, so hold both dates.

From ARPANET to Digital India

1969 - ARPANET, the ancestor of the internet → 1989 - World Wide Web invented by Tim Berners-Lee at CERN → 15 August 1995 - commercial internet reaches India through VSNL → 2003 - NIXI set up to manage the .IN registry → 1 July 2015 - Digital India programme launched

Cloud service models - who manages what

Model	What the provider gives you	What you still manage
IaaS - Infrastructure as a Service	Virtualised servers, storage and network	The operating system and your applications
PaaS - Platform as a Service	A ready development and deployment platform	Only your own application and its data
SaaS - Software as a Service	Finished software delivered over the internet	Nothing - you simply use it

- Deployment models: public cloud, private cloud, hybrid cloud and community cloud.

- MeghRaj is the cloud computing initiative of the Government of India.
- A URL identifies a resource - protocol, domain name and path. A domain name is translated into an IP address by DNS.
- Common domain endings: .com commercial, .org organisation, .gov government, .edu education, .in the India country code.
- A browser (Chrome, Firefox, Edge) displays pages; a search engine (Google, Bing) finds them. They are not the same thing.

The new technologies - where RPSC is now moving

This is the growth area of the chapter. The examiner is not testing whether you can build any of these. It is testing whether you know the one-line definition and the Indian mission attached to it - the year, the outlay and the ministry. So learn the table, then learn the missions table, and stop there.

The technologies - definition plus the fact that gets asked

Technology	One-line definition	Exam anchor
Internet of Things (IoT)	A network of physical objects fitted with sensors that collect and exchange data	Objects plus sensors plus connectivity - not just phones and computers
Big data	Data too large, too fast and too varied for ordinary tools to handle	Five Vs - volume, velocity, variety, veracity, value
Blockchain	A decentralised, immutable ledger distributed across many computers	Bitcoin (2009) attributed to 'Satoshi Nakamoto'; RBI began Digital Rupee (CBDC) pilots in 2022
Artificial intelligence	Machines performing tasks that would need human intelligence	Term coined by John McCarthy at the Dartmouth Conference, 1956; Alan Turing proposed the Turing Test in 1950
Machine learning	Systems that learn patterns from data instead of being explicitly programmed	Supervised (labelled data), unsupervised (clustering), reinforcement (rewards and penalties)

Technology	One-line definition	Exam anchor
Generative AI	Models that produce new text, images, audio or code	Built on the transformer architecture, introduced in 2017
Quantum computing	Computing with qubits instead of bits	Superposition and entanglement; a qubit is the unit of quantum information
5G	Fifth-generation mobile telephony - high speed, low latency	Launched in India on 1 October 2022 at the India Mobile Congress

India's missions in these areas - years, outlays, ministries

Learn the parent department for each - RPSC swaps them

Mission or document	Key numbers	Under
National Strategy for AI, '#AIforAll'	2018	NITI Aayog
IndiaAI Mission	Approved 7 March 2024, about Rs 10,372 crore, seven pillars	MeitY
Bhashini	India's language AI platform for Indian languages	MeitY
National Quantum Mission	Approved 19 April 2023, Rs 6,003.65 crore, period 2023-24 to 2030-31; target 50-1000 physical qubits in eight years	DST
NQM thematic hubs	Computing at IISc Bengaluru; communication at IIT Madras; sensing and metrology at IIT Bombay; materials and devices at IIT Delhi	DST
Bharat 6G Vision	Document March 2023; Bharat 6G Alliance July 2023; leadership targeted around 2030	Department of Telecommunications
National Supercomputing Mission	Launched 2015	MeitY and DST, executed through C-DAC Pune and IISc Bengaluru

- PARAM 8000 (1991), built by C-DAC under Dr Vijay Bhatkar, was India's first indigenous supercomputer.
- PARAM Shivay at IIT (BHU) Varanasi, 2019, was the first system installed under the National Supercomputing Mission.
- Three PARAM Rudra supercomputers were dedicated to the nation in September 2024 - at Pune, Delhi and Kolkata.
- AIRAWAT, at C-DAC Pune, ranked 75th in the Top500 list announced at ISC 2023 - India's AI supercomputing platform.
- Pratyush and Mihir are the supercomputers used for weather and climate forecasting.

CURRENT LINK

Current link, as of September 2026: the National Quantum Mission (2023-24 to 2030-31) and the IndiaAI Mission (March 2024) are both mid-run, so the examinable facts are still the approval year, the outlay and the ministry - Rs 6,003.65 crore under DST for quantum, about Rs 10,372 crore under MeitY for AI. Watch for the four NQM hub institutions being asked as a match; that pattern is already visible in mock and state papers.

Cyber security - malware, agencies and the law

Cyber security questions come in three shapes: identify the malware from its behaviour, name the agency and its ministry, or state a fact about the IT Act or the DPDP Act. The behaviour descriptions are the easiest marks in the chapter because each type of malware does exactly one distinctive thing.

Malware - one behaviour each

Type	How it behaves
Virus	Needs a host programme or file to attach to and spread

Type	How it behaves
Worm	Spreads by itself across a network; needs no host
Trojan horse	Disguises itself as a legitimate programme to get installed
Ransomware	Encrypts the victim's files and demands payment for the decryption key
Spyware and keylogger	Silently collect information and record keystrokes
Botnet	A network of compromised machines controlled remotely by an attacker
Adware	Pushes unwanted advertisements onto the system

- Phishing - a fake email or website that tricks you into giving up passwords or card details. Over a phone call it is vishing; by SMS it is smishing.
- A firewall filters traffic between a trusted internal network and the outside world, allowing or blocking it by rule.
- Encryption converts readable data into cipher text. Symmetric encryption uses one shared key; asymmetric (public key) encryption uses a key pair.
- A digital signature is created with the sender's private key and verified with the sender's public key.
- A digital signature gives three assurances: authenticity of the sender, integrity of the message, and non-repudiation.

Laws and agencies - fix the ministry with the name

Item	Key facts
IT Act, 2000	Based on the UNCITRAL Model Law on Electronic Commerce, 1996; came into force 17 October 2000; amended in 2008
Section 66A	Penalised 'offensive' messages; struck down by the Supreme Court in Shreya Singhal v. Union of India (2015)
CERT-In (2004)	National nodal agency for computer security incidents; under MeitY
NCIIPC	Protects critical information infrastructure; under NTRO, created under Section 70A

Item	Key facts
I4C	Indian Cyber Crime Coordination Centre; under the Ministry of Home Affairs; runs the cybercrime portal and helpline 1930
Cyber Swachhta Kendra	Botnet cleaning and malware analysis centre
DPDP Act, 2023	Data Principal, Data Fiduciary, Consent Manager, Significant Data Fiduciary; sets up the Data Protection Board of India; maximum penalty Rs 250 crore; DPDP Rules notified 14 November 2025

TRAP

Two swaps to guard against. The maximum penalty under the DPDP Act 2023 is Rs 250 crore, not Rs 50 crore. And CERT-In sits under MeitY while I4C sits under the Home Ministry - RPSC exchanges the two ministries in options.

E-governance - India and Rajasthan

This is the highest-yield section of the chapter. Rajasthan e-governance has supplied a question in 2016, 2018, 2021 and 2024. The format is almost always name-to-purpose matching, so build the two tables below and revise them as tables, not as prose. Note the launch dates - they are asked directly.

Digital India and its building blocks

Initiative	Year	What it does
Digital India	1 July 2015	Umbrella programme - three vision areas and nine pillars
MyGov	2014	Citizen engagement platform
DigiLocker	2015	Cloud platform for storing and sharing authentic digital documents

Initiative	Year	What it does
UMANG	2017	Single mobile app giving access to many government services
Aadhaar / UIDAI	UIDAI 2009; Aadhaar Act 2016	12-digit unique identity number
UPI	NPCI, 2016	Instant mobile-based payments
ONDC	2021	Open Network for Digital Commerce
BharatNet and Common Service Centres	-	Rural broadband backbone and last-mile service delivery points
SWAYAM, SWAYAM PRABHA, DIKSHA, NPTEL, e-PathShala, PM eVidya	-	ICT in education - online courses, DTH channels, school content

ASKED IN RAS

Asked in RAS Pre 2016 - which state was declared India's first digital state? Answer: Kerala, in February 2016. RPSC thought so highly of this fact that it asked it TWICE in the same 2016 paper.

TRAP

Kerala, not Rajasthan, is India's first digital state. Rajasthan is genuinely strong on e-governance, and the mind slides towards it - which is precisely why the option is there. Do not fall for it.

Rajasthan - DoIT&C is the nodal department

Rajasthan's portals and schemes - pure matching material

Portal or scheme	Purpose
e-Mitra	Electronic delivery of 600-plus government and private services through 80,000-plus kiosks

Portal or scheme	Purpose
Rajasthan SSO	Single sign-on access to state online services
Jan Aadhaar	Launched 18 December 2019 on the principle 'One Number, One Card, One Identity'; subsumed Bhamashah
Bhamashah Yojana	Relaunched 2014; benefits routed by treating the adult woman as head of the family
Raj Kaj	e-Office and service matters of government employees
RAJ NIVESH	Single-window clearance for investors
iStart	Startup registration, incubation and support
RajNET and RajSWAN	State connectivity backbone
Rajasthan Sampark	Grievance redressal; helpline 181
R-CAT, Jaipur	Rajasthan Centre for Advanced Technology - advanced technology training
RKCL's RS-CIT	Basic digital literacy certification course

ASKED IN RAS

Asked in RAS Pre 2018 - what are e-Mitra kiosks primarily meant for? Answer: electronic delivery of government and private services to citizens. Asked in RAS Pre 2021 - the principle of Jan Aadhaar Yojana? Answer: 'One Number, One Card, One Identity'.

ASKED IN RAS

Asked in RAS Pre 2024 as a four-pair match - Raj Kaj to e-office and employee service matters, RAJ NIVESH to single-window investment clearance, iStart to startups, Rajasthan SSO to single sign-on. Learn these four pairs word for word; they have now been asked in this exact form.

REVISION RECAP

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3. CPU = ALU + Control Unit + registers. Registers are the FASTEST memory, not the slowest.
4. Hierarchy: registers, cache, RAM, secondary storage. RAM volatile, ROM not. EPROM erased by UV, EEPROM electrically, flash is a form of EEPROM.
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14. Cloud: IaaS infrastructure, PaaS platform, SaaS software. MeghRaj is the Government of India cloud.
15. Big data's five Vs; blockchain is decentralised and immutable; Bitcoin 2009 'Satoshi Nakamoto'; RBI Digital Rupee pilots 2022.
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